



Project Catan

ITCS251 Programming in Python



Introduction to Catan

Catan (The Settlers of Catan) is a modern board game first released in 1995.

To emphasize strategy, negotiation, and player interaction. Catan simply simulates basic economic systems and political decision-making, as players must trade, cooperate, and compete, reflecting concepts of resource management, diplomacy, and territorial expansion in early societies.

Original Board Game Interface

Board



Equipments

- Dice x2
- Building Cost Board (roads, settlements, city)
- Colored Building Markers (representing number of players)
- Resource Cards

Overview

STARTING MAP FOR BEGINNERS

To make it as easy as possible for you to get started with *Catan*, we use an award-winning rules system, which consists of 3 parts—the *Overview*, the *Game Rules*, and the *Almanac*.

If you've never played *Catan*, please read the game *Overview* first—it's on the back cover of this booklet. Next, read the *Game Rules* and start to play. And finally, if you have questions during the game, please consult the *Almanac* (it begins on page 6).

Illustration A

Begin the game with the resource cards produced by the settlements marked with white stars. See ☆

Robber

ODDS FOR DICE ROLLS

2 & 12	= 3%
3 & 11	= 6%
4 & 10	= 8%
5 & 9	= 11%
6 & 8	= 14%
7	= 17%

RESOURCE PRODUCTION

 Hills Produce Brick	 Forest Produces Lumber	 Mountains Produce Ore	 Fields Produce Grain	 Pasture Produces Wool	 Desert Produces Nothing
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Catan is a strategy game where players build settlements, roads, and cities

Goal: Reach 10 victory points

Board is made of hex tiles that produce resources:

- Wood, Brick, Sheep, Wheat, Ore
- Each player starts with:
 - 2 settlements and 2 roads

The Five Core Resources

- Brick (Hills)
- Wood (Forest)
- Ore (Mountains)
- Wheat (Fields)
- Wool (Pastures)



Setup :before start

Place 19 hexes, add number tokens, and place the robber on the desert.

Each player starts with 2 settlements and 2 roads.

SET-UP PHASE

Begin the “set-up phase” after you build the game map (see Illustration N and *Set-up, Variable* *).

Everyone chooses a color and takes the corresponding game pieces:

- 5 settlements;
- 4 cities;
- 15 roads; and
- 1 building costs card.



Suggested Beginners' Map Set-up

Illustration N



Illustration E

Turn Structure

On your turn:

1. Roll two dice
2. Collect resources (based on the number rolled)
3. Trade resources (with players or bank)
4. Build or buy



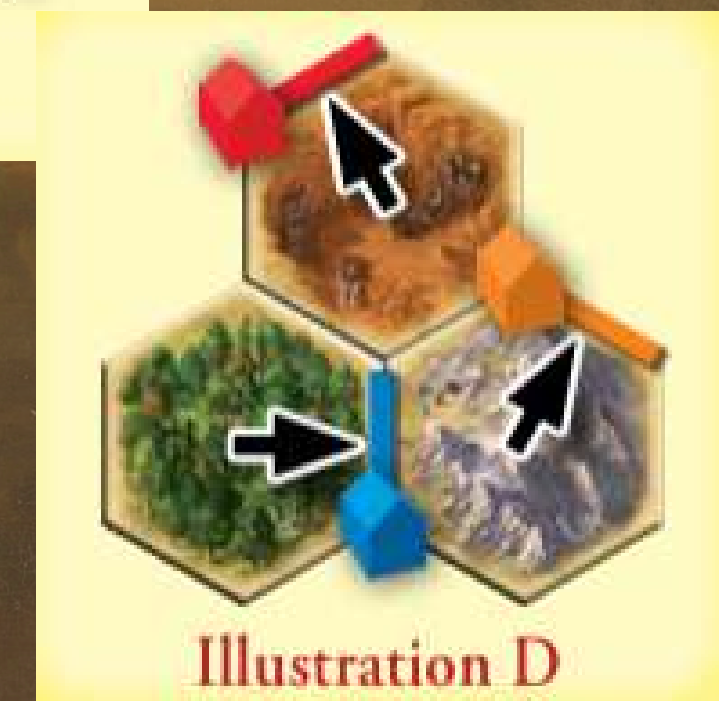
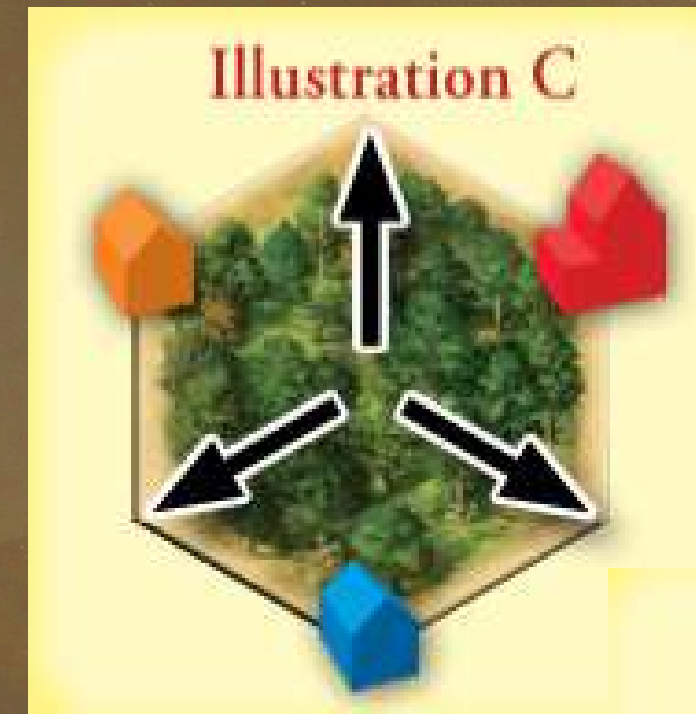
Collecting Resources

If a number rolled matches a tile:

- Players with settlements next to that tile collect resources
- Cities collect 2 resources instead of 1

If a 7 is rolled:

- Robber is activated



Building Costs

- Road:
 - 1 Wood + 1 Brick
- Settlement:
 - 1 Wood + 1 Brick + 1 Sheep + 1 Wheat
- City:
 - 2 Wheat + 3 Ore
- Development Card:
 - 1 Sheep + 1 Wheat + 1 Ore



Trading

- Trade with other players (any deal allowed)
- Trade with the bank:
 - Usually 4:1 ratio
 - Harbors can give better rates

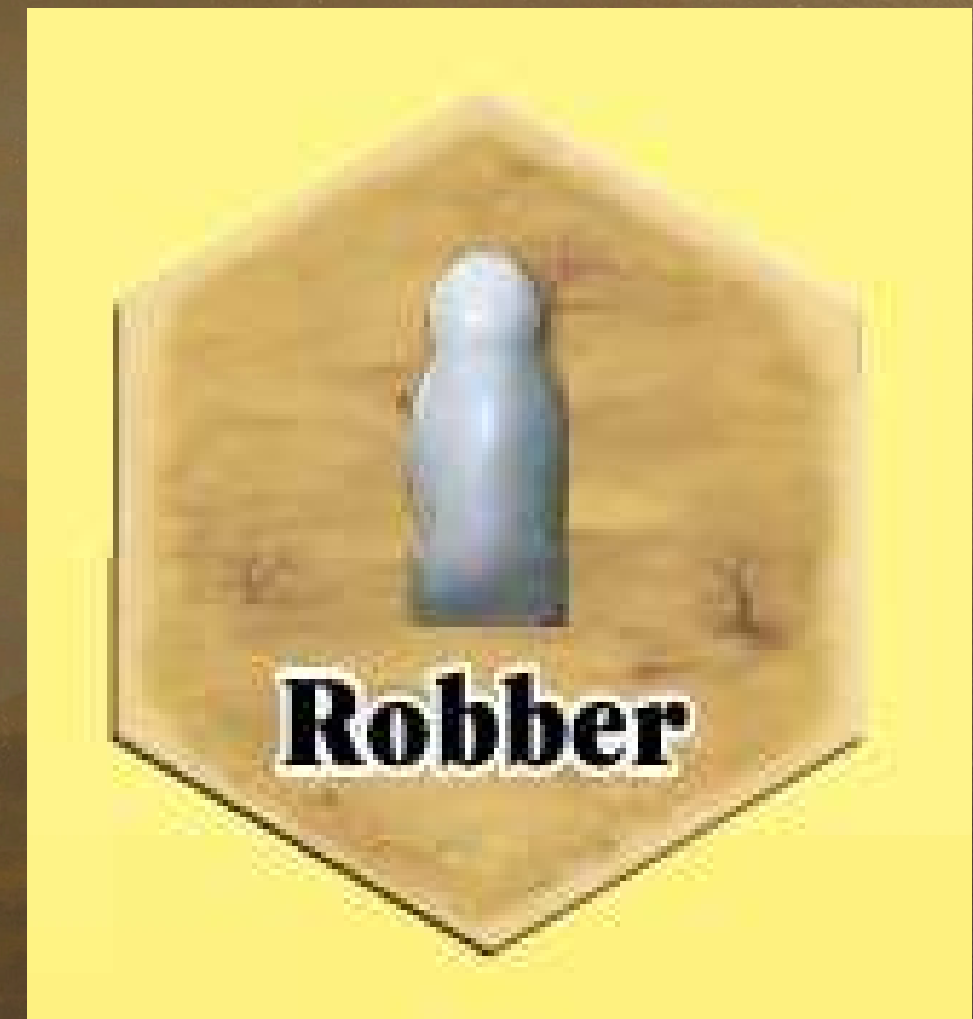


The Robber

Activated when a 7 is rolled

Player must:

- Move the robber to a tile
- Steal 1 card from a player next to that tile
- That tile produces no resources while blocked



Development Cards

Types include:

- Knight (move robber)
- Victory Point
- Road Building
- Monopoly
- Year of Plenty



How to win!

Earn Victory Points:

- Settlement (1), City (2)
- Longest Road (2), Largest Army (2)
- Some Development Cards (1)

Trade smart and expand to win!!!

Goal: Reach 10 victory_points

	= 1 VP	
	= 2 VPs	 
	= 2 VPs	 
	= 2 VPs	 
	= 1 VP	

Prototype

⚓ SETTLERS OF CATAN

A Python Board Game

Number of players:

3

Player names:

Player 1: Alice
Player 2: Bob
Player 3: Carol
Player 4: Dave

▶ Continue Saved Game

🎮 New Game

📜 History

⚓ CATAN

👤 Alice
Phase: Setup Settlement 1
🏆 Victory Points: 0

RESOURCES

🌲 Wood	0
🧱 Brick	0
🐑 Sheep	0
🌾 Wheat	0
⚙️ Ore	0

ACTIONS

- 🎲 Roll Dice
- 🛤️ Build Road
- 🏠 Build Settlement
- 🏰 Build City
- 🏦 Trade with Bank (4:1)
- ✅ End Turn

Save Game Quit

GAME LOG



CATAN

Alice
Phase: Roll Dice
Victory Points: 2

RESOURCES	
Wood	1
Brick	1
Sheep	1
Wheat	0
Ore	0

ACTIONS	
Roll Dice	
Build Road	
Build Settlement	
Build City	
Trade with Bank (4:1)	
End Turn	

Save Game

Quit

GAME LOG

Alice placed a settlement at vertex 43.
Alice placed a settlement at vertex 43.
Alice placed a road on edge (34, 43).



Our Team

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*Thank
You*